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**GE Medical
Systems**

Technical Publication

Direction 2252861-100 FMI Revision 2

GE Medical Systems FMI 25259A DAS Control Board Upgrade

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Revision History

Rev	Date	Reason for Change
Rev. 0	Date: October 8, 1999	Initial Release
Rev. 1	Date: October 26, 1999	Changed return of parts to be non-returned.
Rev. 2	Date: February 27, 2000	Corrected RCIB Connector Numbers (Sect 3, 3.11 & 3.12)

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DAS Control Board Upgrade

EFFECTIVITY:

On all LightSpeed (QX/I) Systems with Gantry Model Number 2180482 that were manufactured prior to July 31, 1999.

PREREQUISITE:

None

PURPOSE:

This FMI is to replace the DAS Control Board (DCB) with a upgraded revision DCB board that supports Z-Axis Tracking that will be available in a future software release.

RELATED FMI'S:

None

MANPOWER REQUIREMENTS:

One Service Engineer 1 hour labor time, plus 2 hours travel time.

SPECIAL TOOLS & TEST EQUIPMENT:

Standard Field Engineer Tool Kit.
Anti-static Wrist Strap & Anti-static Work Mat.

FURNISHED MATERIALS:

FMI kit 25259 - 2253195 consists of the following material and documentation:

Part Number	Description	Qty
2252861-100	FMI 25259 Document	1
2246862	DAS Control Board, Revision G	1

Return of Parts:

Not Required. Any materials should always be disposed of per normal regional/local requirements.

PROCEDURE:

Introduction

The new DCB board revision must be a Rev. G or greater. If not, it will not support Z-axis tracking which will be implemented in a future software release. This FMI is intended to first check the systems current DCB board Revision, and then replace it if necessary.

The FMI is made up of the following sections:

SECTION 1: Check current DCB Revision

SECTION 2: Remove original DCB Board

SECTION 3: Install "new" DCB board

SECTION 4: Re-Test matrix for "new" DCB Board

SECTION 5: Return original DCB

SECTION 1.0 Check current DCB Revision

- 1.1. By using a software development tool called fwInfo, query the DCB and check board revision. If the board is currently a Revision G, then the FMI is not required and the New DCB can be returned as unused.

1.1.1. The computer must be booted and in applications, but Idle (ie. No Scanning).

1.1.2. On the OC, Open a UNIX Shell and **rsh** into the SBC. Type: **rsh sbc** <Enter>

1.1.3. Once logged into the SBC, change directories; **cd /usr/g/bin** <Enter>

1.1.4. At the sbc prompt (ctuser@ct09_sbc}), Type: **fwInfo** <Enter>
(all letter are lowercase, except for the I, which is uppercase)

```
ctuser@ct09_sbc}[3] fwInfo <Enter>
```

```
Enter User fwInfo Request:
```

```
--Ctrl C to exit program
```

```
1 == Full Scan Control Information
```

```
2 == Short Scan Control Information
```

```
.....
```

```
37 == HOP Config Tracking Query DCB
```

```
.....
```

```
99 == Exit
```

```
-> 37 <Return> Select Option 37
```

```
DAS config tracking file will be stored in /usr/g/fw/dcbVer.txt
```

```
-- DAS Control Board --
```

```
Brd: 0 Version: 2138468-01G Alms: 2210116 Ser#:
```

```
-- DAS Converter Board --
```

```
Brd: 1 Version: 2137958-01h Alms: 2210750 Ser#: BI08SK
```

```
.....(List of all 48 Converter boards)
```

```
Brd: 48 Version: 2137958-01h Alms: 2210750 Ser#: BI08VU
```

```
Depress the SPACE key to scroll to bottom of menu.
```

```
-> 99 <Return> to Exit
```

If there is no revision letter, and only a _ follows the brd #, then the board is *not* upgraded.

The "G" at the end of the board number identifies the revision if upgraded.

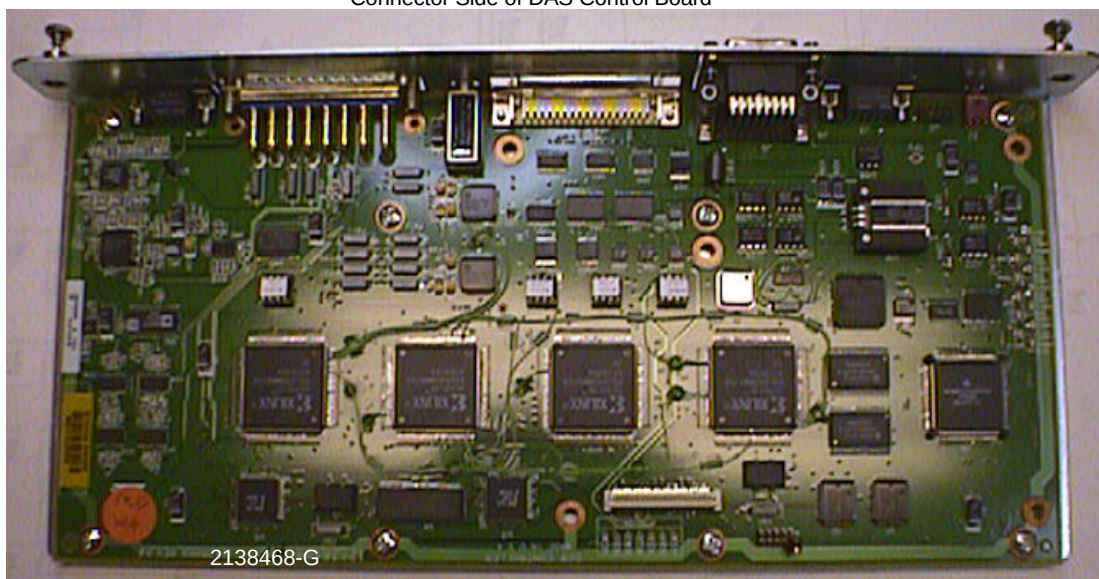
SECTION 2.0 DAS Control Board Removal



NOTICE: Follow ALL Electro-Static Discharge procedures. Always use ESD Wrist Strap and grounding cords. The use of a ground Monitor is suggested

- 2.1. Remove the side covers of the gantry.
- 2.2. Turn off Axial Drive switch and 550 VDC Switch.
- 2.3. Open gantry (Mylar window, side & top covers).
- 2.4. Turn off DAS Power switch. The DAS power switch is behind the center DAS chassis.
- 2.5. Rotate Gantry so that the DCB is at the 3:00 Position (DCB on the service side of the Gantry).
- 2.6. Disconnect Cables going to DCB.
- 2.7. Connector J4; RCIB Cable.
- 2.8. Connector J5; RCIB Terminator. Save this terminator as it needs to be installed on the replacement DCB.
- 2.9. Connector J6; Data cable between Converter Board chassis and DCB.
- 2.10. DAS Data Out Fiber Optic Cable.
- 2.11. Connector J7; DC Power Cable.
- 2.12. Connector J8; KV/mA & Detector Heater Analog signal cable.
- 2.13. Remove 2 screws holding DCB board in DCB chassis.
- 2.14. Remove the DCB and verify Board Revision.
- 2.15. The board revision is found by physical looking at the board and finding the board number near the edge. A new DCB number would read 2138468 - G. If the one being removed does not have the -G designator, it should be replaced. To double check, to the left of the board number is a group of 7 resistor locations (R1 - R7) labeled "Version". The new upgraded board will have R1 and R2 removed.

Connector Side of DAS Control Board



Board P/N & Revision Label

Board Version Resistors

- 2.16. Place the DCB in a anti-static bag.

SECTION 3.0 DAS Control Board Installation



NOTICE: Follow ALL Electro-Static Discharge procedures. Always use ESD Wrist Strap and grounding cords. The use of a ground Monitor is suggested

- 3.1. Remove New DAS Control Board from anti-static bag.
 - 3.2. Verify proper Jumper configuration:
 - 3.3. JP1; No Jumpers for normal operation.
 - 3.4. JP2; No Jumpers for normal operation.
 - 3.5. Align board to card guides in DCB Chassis. Slide the board fully into the chassis and secure using the 2 screws.
 - 3.6. Connect the following cables:
 - 3.7. DAS Data Out Fibre Optic Cable.
 - 3.8. Connector J8; KV/ma & Detector Heater Analog signal cable.
 - 3.9. Connector J7; DC Power Cable.
 - 3.10. Connector J6; Data cable between Converter Board chassis and DCB.
 - 3.11. Connector J4; RCIB Cable.
 - 3.12. Connector J5; RCIB Terminator.
 - 3.13. Connector J3 is not used. There is no cable connection.
 - 3.14. Verify all cables are secure and will not "catch" on anything during rotation.
 - 3.15. Turn DAS Power switch On.
- Note:** The New DCB may fail power-up self-tests due to a firmware mismatch. These messages in the error log can be ignored until after DCB Flash Download has been completed. After the DCB Flash Download, the power-up diagnostic should pass without errors.
- 3.16. Close up Gantry covers.
 - 3.17. Turn on Axial and 550VDC switches.

SECTION 4.0 DCB Re-Test

- 4.1 With applications software up, perform a DCB Flash Download from the Service Desktop.
 From the Service Desktop, Select Replacement Procedures
 DAS Board Change
 Flash Download Tool

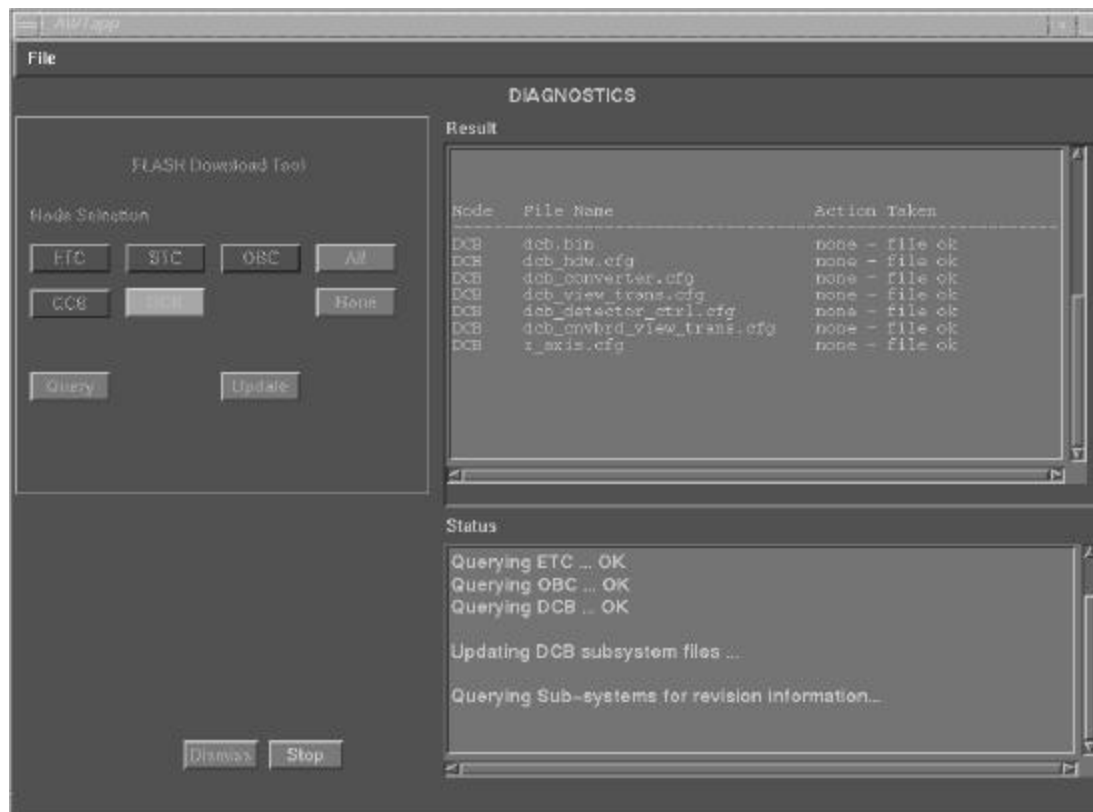


Figure 1 Flash Download Tool

- 4.2 Select Only the DCB (Default has all Nodes selected, ETC,STC,OBC,CCB, &DCB).
- 4.3 Once DCB is only selected, Depress the UpDate option. By viewing the **Status** area, the program will query all the controllers, but only update the DCB. This procedure will take approximately 5 minutes and will report errors in the status error during initial DCB queries. These errors are expected until the final Reset and Query. The sequence of events during this process are:
1. All Controllers are Queried (DCB errors are logged)
 2. Download of DCB.bin file
 3. Controllers are Queried again (DCB errors are logged)
 4. Updates to DCB is performed (6 files)
 5. Controller Resets
 6. Controllers are Queried again (No Errors should occur)

Note: If the Flash Download fails, retry the download.

4.4 Run 10 passes of Scan Data Path Diagnostic.

4.4.1 From the Service Desktop, Select Troubleshooting Menu

Data Acquisition

Scan Datapath Diagnostic

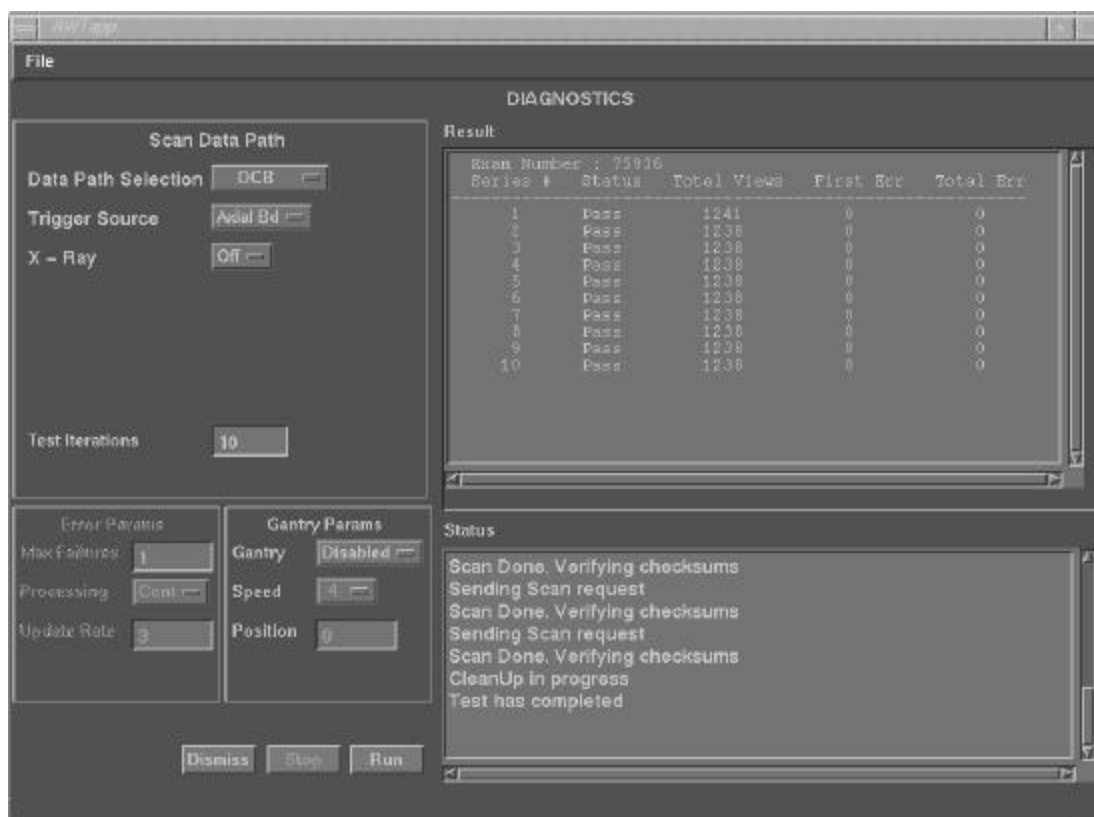


Figure 2 Scan DataPath Diagnostic

4.4.2 From the Main menu, use default options, except change the Test Iterations from 1 to 10, then select "Run". All 10 iterations must pass.

4.5 Take 10 I/Q scans of the 48cm phantom.

Scan Type	Phantom	KV	mA	SFOV	Thickness	Scan Time	Start/End Location	# of Scans
Axial	48cm	120	340	Large	5mm (4i)	1.0 Sec.	17.5 / S7.5	4

The main purpose is to verify that the system scans without any induced DCB errors. Image Quality can be analyzed using the following table with specifications, however the DCB will not cause Image "Means" or other artifacts (Streaks, rings, etc...). If artifacts, troubleshoot using standard operating procedures in the Service System Manual. This troubleshooting would **not** be charged against the FMI.

Image Series Data

Exam / Series/Image <u>Images</u>	AvXc	AvXo	AvXo - AvXc	Avg. Std. Dev.	Comments
1, 5, 9, 13					
2, 6, 10, 14					
3, 7, 11, 15					
4, 8, 12, 16					
Specs	---	---	± 8.5	<40.0	

Box Size: 45 x 45 pixels

Center Coordinates: 256, 256

Outside Coordinates: 256, 60 / 452, 256 / 256, 452 / 60, 256

SECTION 5.0 Dispose of Old DCB

Discard the old DCB according to local/regional procedures and requirements. The DCB can not be used on any other product. It is no longer compatible with LightSpeed Systems with 1.1 Software.

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